Module 11 — EDA & Data Validation

notebooks/eda_validation.py • 337 lines

In one sentence

Proves the generated data actually behaves like real shopping data.

1. What this module does

The generator claims to build in popularity bias, a long tail, sparsity and cold-start cohorts. This module **checks that those properties actually landed**.

That check is not decorative. If the data turned out flat and uniform, every model would be learning from noise and every score reported later would be meaningless. Running ten explicit pass/fail checks means the foundation is verified before anything is built on it.

It also produces the three plots the business case marks as mandatory.

2. Concepts covered

Every idea this module uses, in plain terms.

Concept	In simple terms	Where it is used here
Exploratory data analysis	Look at the data before modelling it	The whole module
Distribution analysis	What shape do the numbers have?	Activity and popularity histograms
Log-log plot	Reveals power laws a normal axis hides	User activity chart
Lorenz curve	Shows how unequally something is shared	Demand concentration
Gini coefficient	One number for that inequality	0.716 here
Sparsity	How empty the matrix is	99.08%
Heatmap	A grid coloured by value	The sparsity visualisation
Data validation	Automated pass/fail checks	The 10 checks at the end
Temporal analysis	How activity changes over time	Monthly volume chart

3. How it works, step by step

1. **Load the three tables** and count them.
2. **Plot user activity** - once on normal axes, once log-log to expose the power-law shape.
3. **Plot item popularity**, the rank-versus-demand curve, and the Lorenz curve.
4. **Compute the Gini coefficient** from the ascending Lorenz curve.
5. **Visualise sparsity** as a heatmap of the densest 60x60 corner - a random sample of a 99% empty matrix would just render as a blank rectangle.
6. **Count the cold-start cohorts** and chart the activity bands.
7. **Chart the funnel** - clicks, carts, purchases - and the rating distribution.
8. **Check the hidden patterns** are recoverable: median price by segment, and how often purchases fall in the declared category.

9. **Run the ten validation checks** and print PASS or FAIL for each.

4. Key functions

Function	What it does
save_figure()	Writes a chart to reports/figures
main()	Runs every check and produces all seven figures

5. Inputs, outputs and how to run

	Detail
Inputs	The three raw tables
Outputs	7 PNG figures in reports/figures, plus 10 pass/fail checks
Result	10 / 10 PASS
Runtime	About 30 seconds
Run with	python notebooks/eda_validation.py

The ten validation checks

Check	Threshold	Observed	Result
Users	at least 5,000	6,000	PASS
Items	at least 2,000	2,500	PASS
Interactions	at least 100,000	138,345	PASS
Sparsity	over 95%	99.08%	PASS
Popularity bias	top 10% hold over 40%	62.9%	PASS
Long tail	Gini over 0.5	0.716	PASS
Cold-start users	present	480 (8.0%)	PASS
Cold-start items	present	255 (10.2%)	PASS
Ratings skew high	mean over 3.5	3.91	PASS
Category preference real	over 30%	60.7%	PASS

7. Things to remember

A real bug this module caught: the Gini came out negative

The first version reported **Gini = -0.716**, which is mathematically impossible. The cause: the Gini formula is defined against the *ascending* Lorenz curve, but the chart plots the conventional descending 'top X% drive Y% of demand' curve, which sits above the diagonal. Integrating that one returns the value with its sign flipped. The magnitude was correct, which is exactly why it was easy to miss. The fix computes the Gini from a separately sorted ascending curve and leaves the chart as it was.

Why sample the densest corner for the heatmap

A random sample of a 99% empty matrix renders as a blank rectangle and shows nothing at all. Taking the 60 most active customers against the 60 most popular products shows the structure - and even that corner is mostly empty, which is the point.

8. Likely questions

Q. Why validate synthetic data you generated yourself?

A. Because the generator could easily fail to produce the properties it intends. The parameters interact - the Zipf offset, the activity distribution, the cold-start carve-outs. Checking is the only way to know the models are being trained on something realistic. It also caught two real calibration problems.

Q. What is the Gini coefficient telling you?

A. How unequally demand is spread. 0 would mean every product sells equally; 1 would mean one product takes everything. At 0.716 there is a strong head and a long tail, which is what real catalogues look like.

Q. What do the mandatory plots show?

A. User activity is a power law - a few very active customers, most barely active. Item popularity is Pareto - the top 10% take 62.9% of demand. And the sparsity heatmap shows that even the busiest corner of the matrix is mostly empty.